

Session 11 (AIML) - Assignment Questions

Q1. (Basic try-except)

Write a program that takes two numbers as input from the user and prints their division.

Use try-except to handle the case when the user enters non-numeric input (ValueError).

Print a friendly message if an error occurs.

Q2. (Handling ZeroDivisionError)

Write a program that divides two numbers entered by the user.

Use try-except to specifically catch ZeroDivisionError and print:

"Error: Division by zero is not allowed."

Also handle ValueError for invalid input.

Q3. (Multiple except Blocks)

Create a program that performs division and also tries to convert a string to integer.

Handle both ValueError and ZeroDivisionError with separate except blocks and appropriate messages.

Q4. (Using else)

Write a program that takes two numbers and divides them.

Use try, except, and else.

The else block should print "Division performed successfully!" only when no exception occurs.

Q5. (Using finally)

Modify the division program to include a finally block that always prints:

"Program execution completed."

Demonstrate that finally runs whether an exception occurs or not.

Add comments explaining the purpose of finally.

Q6. (Combined try-except-else-finally)

Write a complete division program using all four blocks (try, except, else, finally):

- Handle ValueError and ZeroDivisionError.
 - Print success message in else.
 - Always thank the user in finally.
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Q7. (Practical - Age Input)

Write a program that asks the user for their age.

- Convert the input to integer.
 - If the age is negative, raise a ValueError with message "Age cannot be negative."
 - Handle the exception gracefully and print a suitable message.
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Q8. (Multiple Exceptions in One Block)

Create a program that:

- Takes a number as input.
 - Divides 100 by that number.
 - Handle both ValueError and ZeroDivisionError in a **single** except block using a tuple.
Print a combined error message.
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Q9. (Debugging Exception Code)

The following code has issues. Correct it and add comments explaining the fixes:

Python

```
try
    num = int(input("Enter a number: "))
    result = 100 / num
    print("Result:", result)
except ValueError
    print("Please enter a valid number")
```

```
except:  
    print("Some error occurred")
```

Q10. (Mini Project - Exception Handling)

Create a **Simple Calculator** with exception handling:

The program should run in a loop and show this menu:

text

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

- Take user choice and two numbers.
- Perform the operation inside try block.
- Handle ValueError (invalid input) and ZeroDivisionError (for division).
- Use finally to print "Operation attempted." after every calculation.
- Continue until user chooses to exit.